Revisiting SIF abstraction rules with SPARQL for querying BioPAX

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Abstract

BioPAX (Biological Pathway Exchange) is a standard Semantic Web format for the representation of biological pathways. Its expressivity finely describes biological information, but the counterpart is a complexity hindering downstream analyses and reasoning tasks. Exploiting BioPAX' richness while accommodating its complexity is challenging. Abstraction methods on BioPAX partially address this need by extracting contextual information from the knowledge graph. Here, we focus on the abstraction of BioPAX to SIF (Simple Interaction Format), a binary interaction file format modeling interactions between proteins and chemicals. There are few tools for working with BioPAX, and even fewer providing abstractions. Paxtools is the only complete tool capable of handling BioPAX' complexity. With its derivative visualization tool ChiBE, they are the only ones able to abstract BioPAX to SIF using graph patterns based on SIF abstraction rules describing 14 biological configurations. However, SIF rules and Paxtools patterns are ambiguously documented, leading to misunderstandings. In this article we explore SIF rules' formalization through a detailed comparison of their descriptions and Paxtools patterns codes. The result highlights a gap between SIF rules' descriptions and what patterns extract in the graph. We then discuss the potential errors generated by insufficient description of abstraction rules and illustrate them with a biological example. Finally, we propose a transparent and FAIR BioPAX to SIF abstraction method introducing 14 SPARQL queries recapitulating SIF rules. We tested these SPARQL queries by abstracting the PathBank and Reactome human pathway databases in BioPAX to SIF. SPARQL queries and Python code are available at: `https://github.com/CecileBeust/BioPAX-To-SIF-SPARQL.git`

Keywords: BioPAX, SIF, Paxtools, ChiBE, abstraction, knowledge graph

1 Introduction

BioPAX (for Biological PAthway eXchange) is a standard language designed for the representation of metabolic, signaling pathways, molecular and genetic interaction and gene regulation networks [1]. The level 3 of BioPAX was developed in 2010 by a consortium of scientists with the aim of providing a standard format for unifying data representation across different databases, thus improving resources interoperability. BioPAX is a Semantic Web format based on OWL (Ontology Web Language), coming with an ontology providing classes and properties as a controlled vocabulary to finely describe biological processes. The counterpart of BioPAX' rich semantics is its intrinsic complexity, hindering downstream analysis and reasoning tasks. Overall, there is a lack of easy-to-use tools to work with BioPAX data, making its accessibility limited to a range of specialist users. This brings an important need to simplify access to BioPAX and the biological knowledge it contains by overpassing its complexity. In our paper *"BioPAX in 2024: Where we are and where we are heading"* [2] we introduce the need of abstraction methods on BioPAX models. However the only known abstraction of BioPAX has been proposed by Demir et al. in 2013 [3], and allows to abstract BioPAX to SIF (Simple Interaction Format), reducing BioPAX complex mechanistic interactions to simple binary interactions.

The Simple Interaction Format (SIF) was introduced in the early 2000s for describing interactions in a biological network using the Cytoscape network visualization software [4]. SIF is a general binary interaction file format and is adapted for the representation of any type of graph in which edges have properties (that describe pairwise interactions between entities). Within molecular biological networks, entities are proteins and small molecules. SIF rules specify patterns for generating these high-level relations. Note that BioPAX and SIF meet different objectives: BioPAX finely describes complex biological mechanisms, while SIF presents more general patterns and relations between pairs of entities. SIF interactions are presented in tabular format with three columns: a first column for the first entity, a middle column for the interaction type, and a third column for the second entity. This simplicity of representation makes the format easy-to-use and understandable by end-users. Also, some tools require pairwise interaction input formats, like some clustering or module finding algorithms [3], making the use of SIF files relevant in these situations. Moreover, using a format like SIF can be practical for users who want to import their own experimental data [5].

The conversion of BioPAX to SIF can be achieved using specific tools. This conversion is based on inference rules defined by the BioPAX development team and PathwayCommons consortium [8]. A rule specifies a configuration in the BioPAX graph, also named a pattern, and generates the binary SIF interactions corresponding to that pattern.

Paxtools [3] is the only available complete BioPAX API allowing to work with BioPAX data. It is implemented as a Java library and offers several features allowing to manipulate BioPAX data, validate BioPAX files, and also convert files from and to BioPAX. The 14 SIF rules are implemented in Paxtools and modeled with Java graph patterns. Technically, a pattern is defined as a fixed number of BioPAX elements that satisfy a list of constraints [9]. Paxtools patterns work with generative constraints, i.e. successive constraints depend on the previous ones [9]. Paxtools also comes with a pattern search tool, including other patterns than SIF rules, named BioPAX-pattern [9]. While this tool has a GUI interface that minimizes programming efforts, it requires the user to have a fine understanding of the rules and the generative constraints that are applied to the patterns. All Paxtools patterns, including SIF patterns, are available on the GitHub repository of the Paxtools project[1], but are deeply embedded in the Java package, making them complicated to retrieve for non specialist users. In practice, Paxtools patterns can be used to locate and extract specific biological configurations of interest from BioPAX files. Babur et al. [10] recently developed the CausalPath tool which uses 12 patterns from the BioPAX-Pattern tool to extract causal relationships from molecular profiling data and BioPAX pathway data from PathwayCommons.

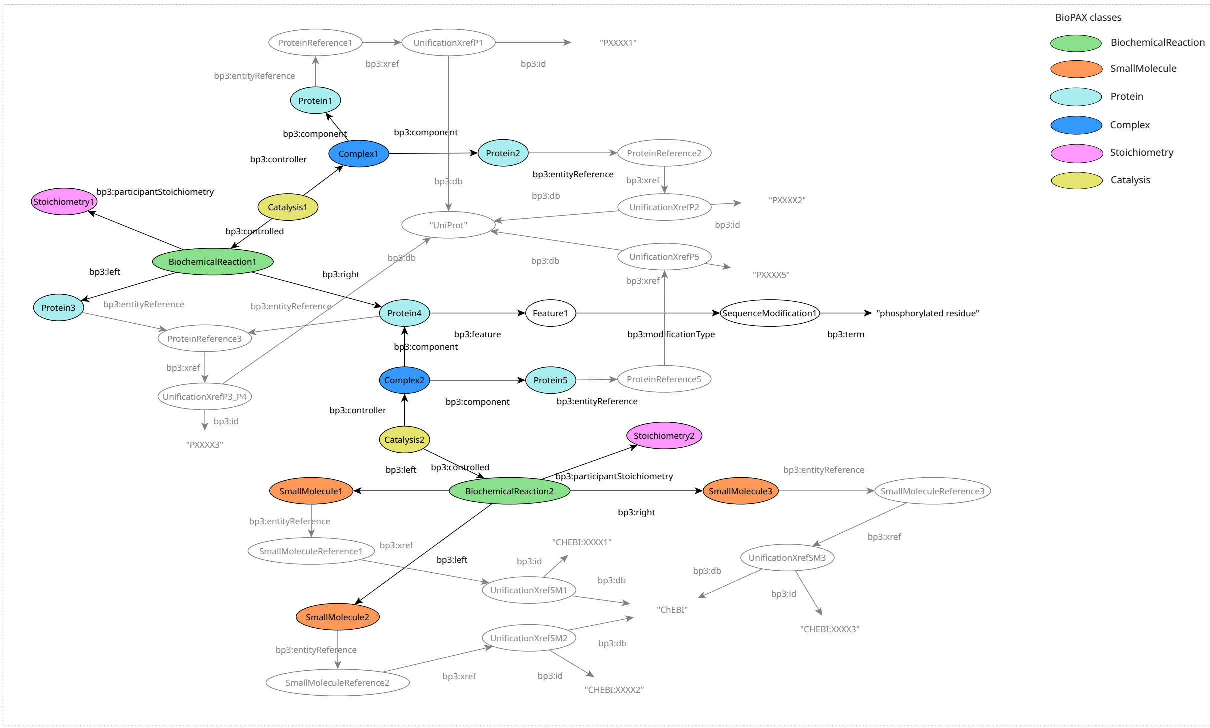
Another solution to convert BioPAX to SIF is to use ChiBE (Chisio BioPAX Editor) [11]. ChiBE is a BioPAX edition and visualization tool based on Paxtools, allowing to generate graphic visualization of BioPAX models using the SBGN Process Description Language [12]. ChiBE supports SIF abstraction of BioPAX models based on encapsulated SIF Paxtools patterns. The resulting SIF abstractions can be visualized in separate windows allowing to work simultaneously on the original BioPAX model and on its SIF abstraction. ChiBE is optimal for small BioPAX files conversion and provides a very good graphic interface to work with BioPAX models, but it does not support the abstraction of larger models like pathway database BioPAX exports.

Although Paxtools and ChiBE are valuable tools, and the only ones offering such a large scope of features to work with BioPAX data, understanding Paxtools patterns remains restricted to specialists or users with programming skills. Moreover, digging into Paxtools patterns codes suggests that the patterns are more complex than the SIF diagrams presented to the users, leading to some ambiguity in the definition of SIF abstraction rules.

In a recent study, Chang et al. [13] used a format similar to SIF named SIFI (for Simple Interaction Format with Intermediates) to integrate ten different knowledge bases from PathwayCommons in a global integrative network named GINv2.0. The main difference with the original SIF format is that intermediate nodes are added to take into account intermediate states of biochemical reactions. The conversion of BioPAX to SIFI is performed using rBiopaxParser [14] which is another wrapper for BioPAX, and is not based on Paxtools patterns nor SIF rules (low-level abstraction). Using this integrative network, the authors jointly explore metabolic and signaling pathways integrated from different databases with distinct scopes and unified in a SIF-based format, showcasing the added

[1] https://github.com/BioPAX/Paxtools/blob/master/pattern/src/main/java/org/biopax/paxtools/pattern/PatternBox.java

A) BioPAX graph



Extraction of SIF relations from BioPAX

B) SIF tabular file

Left	Middle	Right
Protein1	controls-state-change-of	Protein3
Protein2	controls-state-change-of	Protein3
Protein1	controls-phosphorylation-of	Protein3
Protein2	controls-phosphorylation-of	Protein3
Protein4	controls-production-of	SmallMolecule3
Protein5	controls-production-of	SmallMolecule3
SmallMolecule1	consumption-controlled-by	Protein4
SmallMolecule1	consumption-controlled-by	Protein5
SmallMolecule2	consumption-controlled-by	Protein4
SmallMolecule2	consumption-controlled-by	Protein5

C) SIF graph

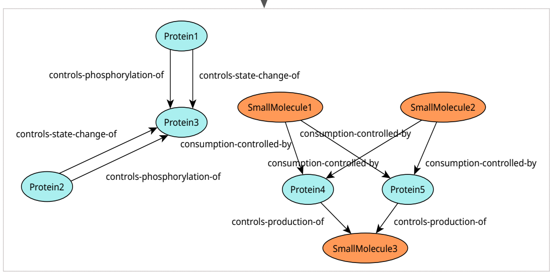


Figure 1: **Conversion of a BioPAX graph to SIF**. A) Toy example of a BioPAX graph. BioPAX entities are nodes and BioPAX properties are edges. Two successive biochemical reactions are presented as green nodes, each one involving proteins and/or small molecules as reactants, products or controllers. Proteins are represented as light blue nodes, complexes are represented as dark blue nodes, catalyses are represented as yellow nodes, stoichiometric parameters are represented as pink nodes and small molecules are represented as orange nodes. Moreover, proteins and small molecules can be gathered in families, which are modeled in BioPAX by linking entities to instances of `bp3:EntityReference` (used to map to external resources like UniProt [6] and ChEBI [7]). B) The resulting SIF tabular file after BioPAX to SIF conversion. The right and left columns contain entities that are linked with a relationship presented in the middle column. C) The visualization of the SIF tabular file is a directed graph. We retrieve entity nodes (proteins and small molecules), that are linked by SIF relationships including: `controls-state-change-of`, `controls-phosphorylation-of`, `controls-production-of` and `consumption-controlled-by`.

value of this format in integrating heterogeneous biological data.

Figure 1 illustrates the conversion of a toy example BioPAX graph into SIF. The BioPAX input and SIF outputs are detailed in different panels. Panel A presents a toy example of two successive biochemical reactions in BioPAX (green nodes). In this graph, nodes represent instances of some BioPAX classes (like `bp3:BiochemicalReaction`, `bp3:Complex`, `bp3:Protein`), and edges represent the BioPAX properties applied between nodes. The participants of the biochemical reactions are detailed. We see both aspects of biological information conveyed by BioPAX: the fine grain of the details and the associated complexity of the representation. Thus, BioPAX complexity is high, even for small models. Abstracting BioPAX to smaller graphs focused on a limited range of entities like SIF can be informative. Here, using SIF abstraction rules, the conversion of this BioPAX graph to SIF generates a SIF tabular file presented in panel B. The output SIF file has three columns: the right and left columns contain entities (source and target nodes) that are linked with a relationship described in the middle column. This SIF file can be visualized as a graph (for example using a network visualization software like Cytoscape), presented in panel C.

In this article, we investigate the granularity of different descriptions of SIF rules, by a detailed study of SIF rules and Paxtools patterns. We show inconsistencies between SIF rule diagrams, their function descriptions in the Paxtools code, and the raw code of the associated Paxtools patterns. We then illustrate how these inconsistencies can lead to potential misunderstandings or interpretation errors of the abstractions for the users. Finally, we propose transparent SPARQL queries based on the 14 SIF abstraction rules available on PathwayCommons, that could complement the BioPAX to SIF abstraction proposed by Paxtools. SPARQL queries are integrated in a Python code within Jupyter notebooks available on GitHub[2].

2 Method

2.1 Definition of the three meanings of description of SIF rules

In order to formally describe the conversion of BioPAX data to SIF, the SIF development team have defined inference rules that define how specific biological configurations in BioPAX should be abstracted to SIF. We will refer to these inference rules as SIF rules in this article. The complete set of SIF rules has not been explicitly published. Technically, SIF rules are implemented as graph patterns encapsulated in the Paxtools Java package. We will call them Paxtools patterns. We compare the different ways SIF abstraction rules are described and we define three meanings of descriptions.

- **Meaning 1** corresponds to the SIF rules description available as diagrams and associated textual descriptions on the PathwayCommons website, in the formats section[3]. There are 14 SIF rules describing the reduction of mechanistic interactions from BioPAX graphs in binary SIF interactions between proteins and/or small molecules. Each rule corresponds to a specific biological configuration in the BioPAX graph, and defines how this configuration is abstracted in a SIF simple binary interaction. Table 1 presents the 14 SIF abstraction rules as they are listed on the PathwayCommons website, as well as their associated textual descriptions that detail the biological configurations associated to each pattern.
- **Meaning 2** corresponds to the Paxtools patterns function descriptions of SIF rules found in the Paxtools code. These descriptions are in the form of comments in the code and describe what the Paxtools patterns associated to SIF rules do.
- **Meaning 3** corresponds to the raw Paxtools patterns codes.

[2]`https://github.com/CecileBeust/BioPAX-To-SIF-SPARQL.git`
[3]`https://www.pathwaycommons.org/pc2/formats`

SIF abstraction rule	Description from PathwayCommons
controls-state-change-of	First protein controls a reaction that changes the state of the second protein
controls-transport-of	First protein controls a reaction that changes the cellular location of the second protein
controls-phosphorylation-of	First protein controls a reaction that changes the phosphorylation status of the second protein
controls-expression-of	First protein controls a conversion or a template reaction that changes expression of the second protein
catalysis-precedes	First protein controls a reaction whose output molecule is input to another reaction controled by the second protein
in-complex-with	Proteins are members of the same complex
interacts-with	Proteins are participants of the same MolecularInteraction
neighbor-of	Proteins are participants or controlers of the same interaction
consumption-controled-by	The small molecule is consumed by a reaction that is controled by a protein
controls-production	The protein controls a reaction of which the small molecule is an output
controls-transport-of-chemical	The protein controls a reaction that changes cellular location of the small molecule
chemical-affects	A small molecule has an effect on the protein state
reacts-with	Small molecules are inputs to a biochemical reaction
used-to-produce	A reaction consumes a small molecule to produce another small molecule

Table 1: **The 14 SIF abstraction rules from PathwayCommons**. The rules are listed on the PathwayCommons website format page[6]. The left column presents the SIF rules names and the right column presents their associated textual description.

For meanings 2 and 3, function descriptions and patterns codes can be found in the GitHub repository of Paxtools[4].

2.2 Comparison of the impact of different SIF rule descriptions and detailed analysis for one SIF rule

In order to compare the impact of different SIF rule descriptions, we first selected one representative SIF rule as an example to detail. We selected the SIF rule `controls-transport-of-chemical` because it is a rule involving both proteins and small molecules. We then developed three SPARQL queries corresponding to the three meanings of description of this SIF rule. We executed the three SPARQL queries on a small BioPAX pathway example, the pathway "Kidney function - Collecting duct" (`SMP0121011`) from PathBank. We then compared the number of SIF interactions obtained by each of the three SPARQL queries in order to choose the appropriate meaning of description. We also compared the results of the three SPARQL queries to results obtained with state of the art tools (Paxtools and ChiBE).

2.3 Generalization of SPARQL queries for BioPAX to SIF abstraction on the appropriate meaning of description

We provide 14 SPARQL queries for the BioPAX to SIF abstraction. For consistency reasons, we have decided to base all our queries on the same meaning of description. Following the experiment described in the Methods section 2.2, we chose the meaning 2 description (Paxtools patterns function descriptions) as a reference meaning for the

[4] `https://github.com/BioPAX/Paxtools/blob/master/pattern/src/main/java/org/biopax/paxtools/pattern/PatternBox.java`

development of queries because it seemed like a good compromise in terms of pattern complexity (neither too general like meaning 1 but nor too complex like meaning 3).

We executed the 14 queries on the BioPAX exports of two pathway databases, PathBank [15] and Reactome [16], to generate their SIF abstraction. For PathBank, we downloaded the BioPAX exports of PathBank human pathways from PathwayCommons (2689 BioPAX files for 2689 human pathways, last updated 2019). For Reactome, we downloaded the latest BioPAX export of the database from the Reactome website[7] (version 94, September 2025) and worked on the Human BioPAX file. To run the SPARQL queries, we loaded the BioPAX files into a local Apache Jena Fuseki SPARQL endpoint with 16Go RAM. SPARQL queries are available in Jupyter notebooks on GitHub [8].

By running the Jupyter notebook using the BioPAX files of the two databases as inputs, the 14 independent SIF SPARQL queries are run and 14 SIF file outputs are generated for each database. Merging the 14 SIF files into a single SIF abstraction for each database was performed using the Cytoscape tool [4].

3 Results

3.1 The study of SIF abstraction rules reveals ambiguity surrounding their definition

We compared the three meanings (SIF rule description from PathwayCommons, Paxtools patterns function descriptions and Paxtools patterns codes) of SIF rules specifications using the SIF rule `controls-transport-of-chemical` as an example (Fig. 2). By converting each description into a UML diagram, we reveal the significant differences in the identified meanings of SIF rule descriptions (Fig. 2, right panel).

Meaning 1 presents the configuration of the SIF rule `controls-transport-of-chemical` as shown in PathwayCommons. It states that the `controls-transport-of-chemical` rule should be applied to every BioPAX configuration where a conversion reaction has a small molecule input (X) that is exported from a first location (location 1) to a second location (location 2). This reaction is controlled by a protein (protein A). The resulting SIF binary interaction is a directed `controls-transport-of-chemical` relationship between protein A and small molecule X. By digging into the Paxtools code, we retrieve the function description of the `controls-transport-of-chemical` pattern, that is presented in the meaning 2. This description states that the SIF relationship should be added at the scale of the `ProteinReference` of the controlling protein. In BioPAX, entities can be linked to references (instances of `bp3:EntityReference`) that are used to store information about a set of related entities (for example families of proteins, in our case instances of the same protein in different locations), and to map to external resources like UniProt [6] or ChEBI [7]. This difference of scale was not specified in the meaning 1 description. Finally, we analyzed the Paxtools pattern code associated to the `controls-transport-of-chemical` SIF rule as a meaning 3 description. In the pattern, some constraints are again added as compared to meanings 1 and 2. For example, the code specifies that the pattern is applied only for small molecules which are not part of a blacklist. This blacklist is a list of ubiquitous small molecules defined by Paxtools, that are considered irrelevant and noisy in SIF interactions (like water and ATP molecules for example). In addition, it is mentioned in the code that the SIF relationship should be added also at the scale of the `SmallMoleculeReference` of the transported small molecule. Another constraint is that the controlling proteins can be part of complexes. These added constraints, found at the scale of the Paxtools code, are thus hidden for the users who won't dig into the Paxtools GitHub repository.

The understanding of SIF rules and the precision of the patterns is thus not straightforward. There is no clear consensus on the definition of SIF rules, and exact graph patterns that should be queried on the BioPAX graph to obtain SIF interactions. SIF diagrams available on PathwayCommons are over simplified as compared to Paxtools patterns and their associated constraints that are complex and highly encapsulated.

[7] `https://reactome.org/download-data`
[8] `https://github.com/CecileBeust/BioPAX-To-SIF-SPARQL.git`

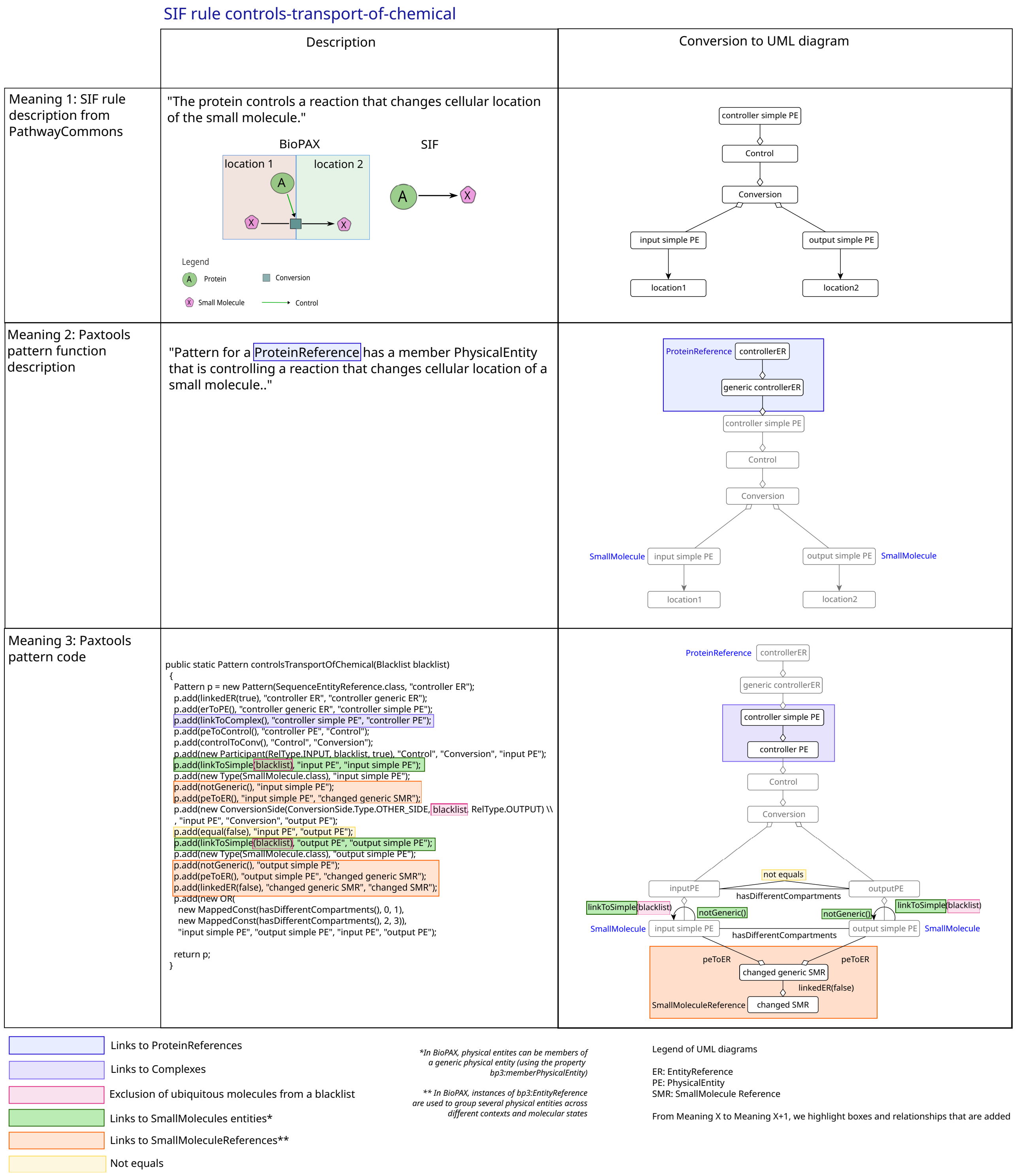


Figure 2: **Comparison of the SIF rule diagram `controls-transport-of-chemical` from PathwayCommmons, its Paxtools pattern function description, and the corresponding Paxtools pattern code**. The left column of the table presents the three meanings of description of the SIF rule, the middle column presents the three descriptions, and the right column presents the conversion of the three descriptions into a UML diagram. The first row represents the meaning 1 of SIF rule description: the diagram representing the SIF rule `controls-transport-of-chemical` and its associated textual description from the PathwayCommons website. The second row presents the meaning 2 of SIF rule description: the Paxtools pattern function description for the `controls-transport-of-chemical` rule as it is written on the Paxtools code. The third row represents the meaning 3 of SIF rule description: the Paxtools pattern code of the SIF rule `controls-transport-of-chemical`. UML diagram legend: `ER: EntityReference`, `PE: PhysicalEntity`, `SMR: SmallMoleculeReference`. From on meaning to the next, we highlight the boxes and relationships that are added.

3.2 Quantification of SIF abstraction rules ambiguity

The study of SIF abstraction rules revealed several meanings of precision of the rule descriptions, whether at the scale of diagrams or at the scale of the Paxtools patterns function descriptions and code. These differences in rule descriptions are not specified to the users, and they could lead to different abstraction results depending on the meaning of description we are looking at. We wanted to investigate and quantify the impact of these different granularities of SIF rule description when abstracting the same pathway with the constraints of the three meanings of description. To this end, we used SPARQL queries as a transparent way to compare the different meanings of description. We coded three SPARQL queries that capture the constraints of the three meanings of SIF rules descriptions and generated the resulting SIF abstractions using a small pathway example in BioPAX. For each one of the three meanings of descriptions, we compared the number of SIF interactions obtained. Also, we compared the number of interactions obtained with the three SPARQL queries to results obtained with state of the art tools (Paxtools and ChiBE).

Figure 3 presents the comparison of abstractions of a PathBank pathway using the SIF rule `controls-transport-of-chemical` for the three meanings of description. The meaning 1 SPARQL query searches in the BioPAX graph a configuration where a reaction translocates a small molecule from one cellular location to another, and where a protein directly controls this reaction. This meaning 1 SPARQL query resulted in 14 SIF interactions. The meaning 2 SPARQL query captures the results of the meaning 1 but adds a new constraint found in the patterns function description, that is that the controller protein should be reached by the intermediate of its `EntityReference`. This meaning 2 SPARQL query resulted in 5 SIF interactions. It makes sense that we get back fewer results than with the meaning 1 query, because here we exclude four protein controllers that are not linked to a reference in BioPAX. Last, the meaning 3 SPARQL query gathers constraints found directly in the Paxtools patterns code of the SIF rule `controls-transport-of-chemical`. The first added constraint is that the instances of `bp3:SmallMolecule` at the left and at the right of the reaction should be linked to the same instance of `EntityReference`. It makes sense because we are looking for a configuration where a protein is translocated, and this is described with two distinct nodes, representing two different states of the protein (in two different cellular locations), linked to the same reference in BioPAX. Also, the controlling protein can be located inside a complex, and we have to unfold the complex to retrieve its protein components. Another added constraint is that the translocated small molecule should not be part of a blacklist of ubiquitous small molecules defined in the query (in meaning 3 results, interactions involving `H2O` and `H+` molecules are removed because of the Paxtools blacklist). The meaning 3 SPARQL query resulted in 24 SIF interactions with the pathway test, which is much more than with the previous queries. This is logical since we get back protein controllers located in controlling complexes, that we did not extract with the meaning 1 and meaning 2 queries.

In Figure 3D, we compared these results with results obtained with state of the art tools, that are Paxtools (command line SIF converter and graphical interface of BioPAX-pattern) and the SIF converter of ChiBE. ChiBE yielded 27 interactions on the pathway example. These are the same ones than the meaning 3 interactions but with the three additional blacklisted interactions involving `H2O` and `H+` molecules (ChiBE does not use an integrated blacklist of ubiquitous molecules). It shows that our meaning 3 SPARQL query captures well the constraints of the Paxtools pattern for this SIF rule. However, using the BioPAX-pattern graphical interface, we only get 25 results. Interactions involving the protein `ATP1B4` are missing for an unknown reason, and one blacklisted interaction (involving `H+`) is not removed. This may be due a to the use of different versions of Paxtools patterns between BioPAX-pattern and ChiBE. And last, the command line SIF converter of Paxtools yields no results, which we are unable to explain. A hypothesis is that the Paxtools command line tool was unable to fetch the identifiers of proteins and/or small molecules because of mapping issues (private correspondence with the Paxtools authors).

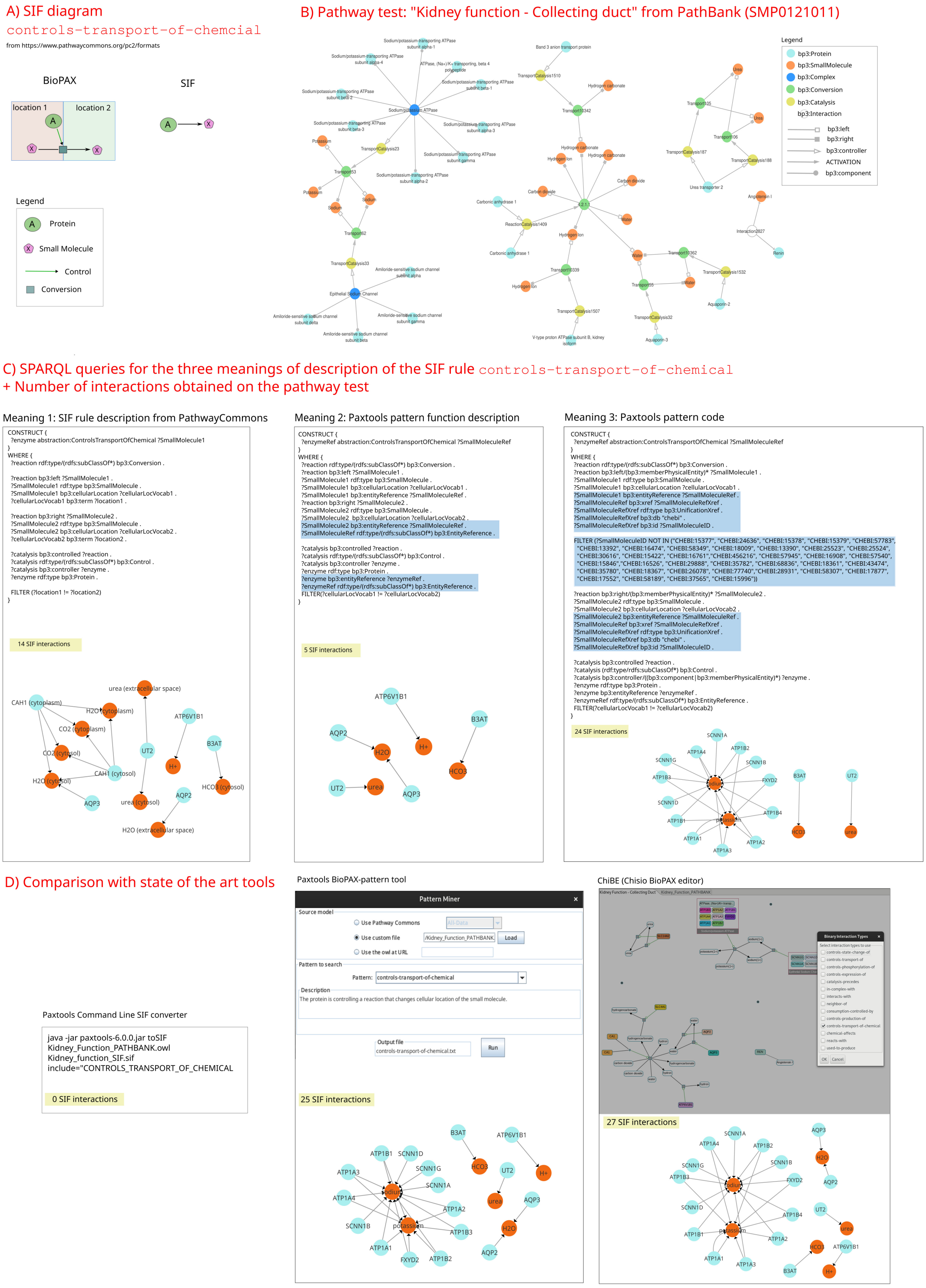


Figure 3: **Impact of SIF rule description ambiguity on the BioPAX to SIF abstraction**. Panel A presents the diagram of the SIF rule `controls-transport-of-chemical` which is used as an example here. Panel B presents a simplified visualisation (using Cytoscape) of a BioPAX [9] pathway from PathBank: "Kidney function - Collecting duct" (`SMP0121011`). Panel C presents three SPARQL queries that correspond to the three meanings of description of the SIF rule `controls-transport-of-chemical`. When going from meaning X to meaning X+1, the added constraints are highlighted in blue in the corresponding SPARQL queries. Panel D presents the results obtained on the same pathway test from panel B using state of the art tools Paxtools command line SIF converter, Paxtools BioPAX-pattern and ChiBE. In panel C and D, the number of SIF interactions obtained with each SPARQL query or each tool on the pathway example of panel B is indicated, as well as the corresponding SIF interaction networks where proteins are represented as light blue nodes and small molecules as orange nodes.

3.3 Generalization of SPARQL queries for BioPAX to SIF abstraction

The study of SIF abstraction rules and associated Paxtools patterns revealed different granularities of rules descriptions, that could lead to some ambiguity for the users. In order to provide a transparent way to abstract BioPAX to SIF, we wrote 14 SPARQL queries that model the 14 SIF rules and their associated constraints. We chose SPARQL for its transparency and its compliance with Semantic Web and FAIR principles.

When writing SPARQL queries, we had to choose which meaning of description and which constraints to apply for every SIF rule. We decided that meaning 1 was not precise enough, and that meaning 3 was, on the contrary, too complex to represent on a single query. We thus designed our SPARQL queries to match the meaning 2 description of SIF rules from the patterns function descriptions in Paxtools, as a compromise in terms of complexity. Unfortunately, for some queries the Paxtools patterns function description was missing or incomplete in the Paxtools code, and we had to write or complete ourselves a meaning 2 for these queries consistently from one query to another. We called these descriptions "explained meaning 2", and based the development of our 14 SPARQL queries on it (Supplementary File S-1). We only explained the meaning 2 when the pattern description found in Paxtools code was too short or not detailed enough, making it inconsistent with the other meaning 2 descriptions. We wrote a SPARQL query like the one presented in Figure 4 for each one of the 14 SIF abstraction rules. The 14 SPARQL queries are available on GitHub[9]. The 14 queries are also recapitulated with their textual descriptions (meaning 1, meaning 2 and explained meaning 2) in a document available as a Supplementary file S-1[10].

To illustrate the generalization of a SIF rule into a SPARQL query, we detail the SPARQL query of the SIF rule `controls-transport-of-chemical` (Figure 4). The explained meaning 2 is used to write the SPARQL query of the `controls-transport-of-chemical` SIF rule presented on the right side of the figure. The SPARQL query is a `CONSTRUCT` query, meaning that it generates a new graph from the BioPAX input data, following the constraints of the explained meaning 2 recapitulated in the query. The generated graph is a SIF abstraction with interactions between a `ProteinReference` and a `SmallMoleculeReference` in a translocation event. The predicate of these interactions is an abstraction relationship `abstraction:ControlsTransportOfChemical` that we defined specifically for the SIF abstraction. For this example, we also provide a diagram recapitulating the constraints in the SPARQL query, in order to provide a visual indicator of the constraints of the SPARQL query. The diagram shows the BioPAX graph pattern that is extracted with the query to create the SIF interaction between the SPARQL variables `?enzymeRef` and `?SmallMoleculeRef`.

3.4 Application of SPARQL queries to whole pathway databases

Use of SPARQL to represent SIF queries aim to reconcile the heterogeneity of SIF rules descriptions for avoiding ambiguities. To demonstrate its efficiency, we tested our 14 SPARQL queries on the BioPAX exports of large pathway databases, PathBank [15] and Reactome [16]. For each database, we executed the 14 SPARQL queries independently on the BioPAX export of the database, generating 14 independant SIF views that can be merged to create a unique SIF abstraction of the two databases.

Table 2 and Table 3 summarize the results obtained for the SIF abstraction of PathBank and Reactome human BioPAX pathway data respectively. In each case and for each SIF abstraction, we detailed the number of nodes and edges obtained, as well as the proportion of edges of each type in the global merged SIF abstraction network. For PathBank, 76.6% of edges in the SIF abstraction come from the `catalysis-precedes` rule, while other rules are less represented such as `in-complex-with` (6%), `consumption-controlled-by` (4.9%), `controls-production` (5.1%) and `used-to-produce` (4.7%). For Reactome, the majority of SIF edges comes from the `in-complex-with` rule with 32% of edges, other edge types being mainly `catalysis-precedes` (29%), `neighbor-of` (15.1%) and `used-to-produce`

[9] https://github.com/CecileBeust/BioPAX-To-SIF-SPARQL.git

[10] https://github.com/CecileBeust/BioPAX-To-SIF-SPARQL/blob/main/QueriesDocumentation/SupplementaryMaterialS-1.pdf

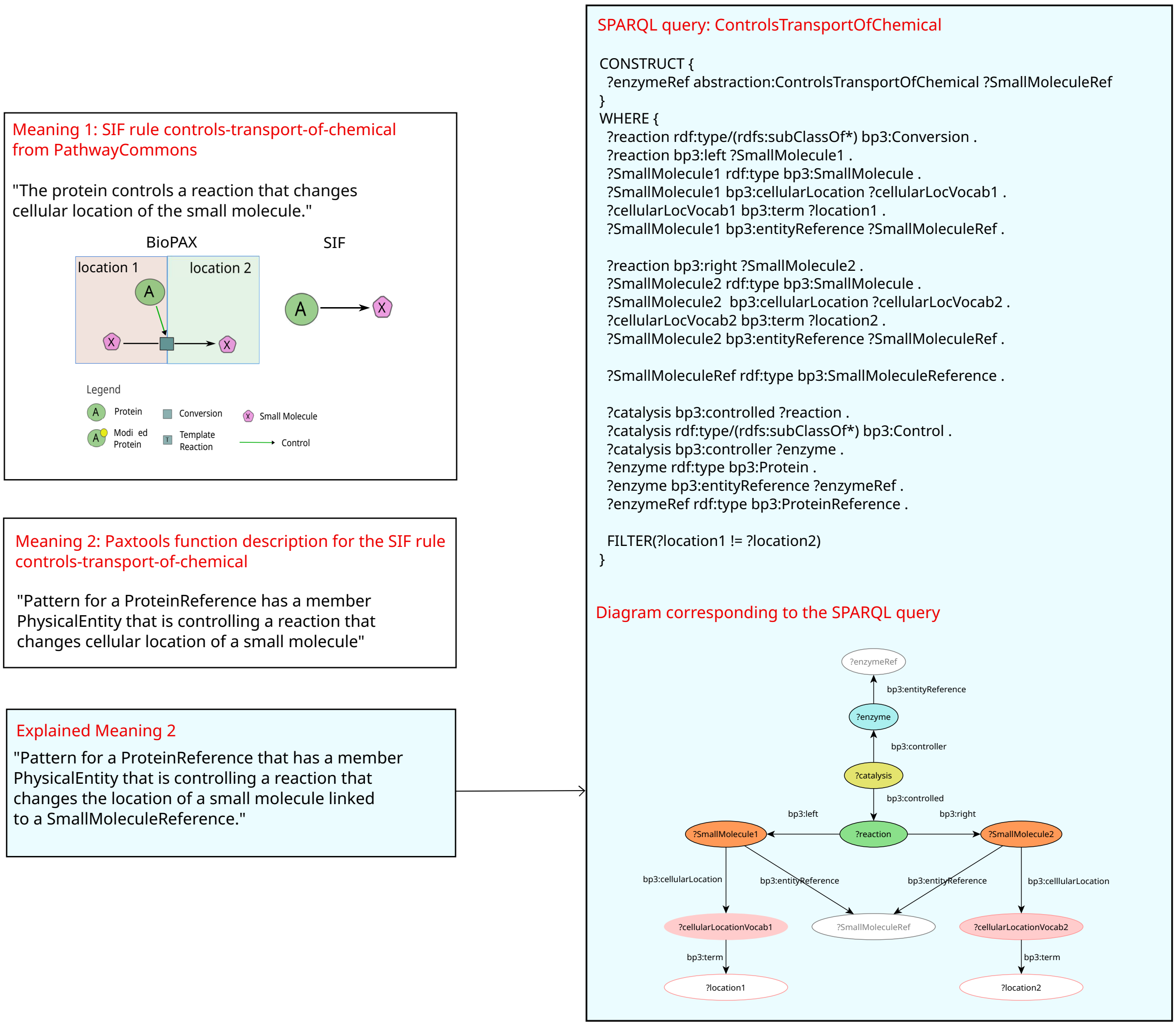


Figure 4: **SPARQL representation of a SIF rule**. The left panel presents the meaning 1 (SIF diagram and text from PathwayCommons), meaning 2 (Paxtools patterns description) and explained meaning 2 descriptions of the SIF rule `controls-transport-of-chemical`. The explained meaning 2 details in a consistent manner the sentence of meaning 2, so that the user clearly understands that the SIF rule applies between two instances of `bp3:EntityReference` (a `ProteinReference` that controls the transport of a `SmallMoleculeReference`. The explained meaning 2 is used to write the SPARQL query of the `controls-transport-of-chemical` SIF rule presented in the right panel. The corresponding visualization of the query's constraints is also presented as a diagram in the right panel.

SIF abstraction rule	**Number of nodes**	**Number of edges**	**Proportion of edges (%)**	**Execution time (s)**
controls-state-change-of	8	4	0.002	0.57
controls-transport-of	76	52	0.02	0.09
controls-phosphorylation-of	0	0	0	0.14
controls-expression-of	0	0	0	0.07
catalysis-precedes	3176	164702	75	594.04
in-complex-with	4044	13086	6	0.64
interacts-with	548	610	0.3	0.09
neighbor-of	1247	1859	0.85	0.48
consumption-controlled-by	4943	10677	4.9	0.77
controls-production	5167	11214	5.1	1.03
controls-transport-of-chemical	526	692	0.31	0.17
chemical-affects	42	39	0.02	0.06
reacts-with	1657	6396	2.9	0.52
used-to-produce	2667	10316	4.7	211.39
Global SIF abstraction of PathBank	10500	219647		

Table 2: **Number of nodes and edges obatained in the SIF abstraction of PathBank (Homo Sapiens, version 2019 from PathwayCommons) generated using SPARQL queries**. For each SIF abstraction rule, the number of nodes and edges are presented, as well as the proportion of SIF edge types with regard to the global aggregated SIF network. The total number of nodes and edges for the global SIF abstraction merging the 14 SIF abstractions is also precised.

(9.9%). These different orders of magnitude observed between the obtained SIF interactions for the two databases reflect their differences in scopes. Some SIF rules do not generate results, such as the `controls-phosphorylation-of` and `controls-expression-of` rules for PathBank. In this case, we do not clearly know if this absence of results is due to a difference in the encoding of reactions in the BioPAX export of PathBank that do not match the rule, or if phosphorylation and expression events are not described at all in the PathBank database (but this is out of scope for this article). We also observe that we do not obtain results for the SIF rule `interacts-with` with the BioPAX export of Reactome. In this case, this is because Reactome do not make use of the `bp3:MolecularInteraction` BioPAX class for which interactions are captured with the `interacts-with` rule.

Overall, our SPARQL queries scales for generating SIF views from BioPAX data on large databases like PathBank and Reactome. Moreover, our queries are fast even on these large files, the execution of queries on PathBank and Reactome was on average few seconds or less than a minute per SPARQL query (maximum 4-5 minutes for the `chemical-affects` query on Reactome). The generated SIF abstraction can be analyzed either independently or conjointly aggregated within a merged SIF abstraction network. However, users should keep in mind the heterogeneity of SIF edge types proportions in the merged abstraction, with some edge types being over-representated (mainly `in-complex-with` and `catalysis-precedes`) which may affect the graph topology. The SIF rules selected by the users should depend on the entities and/or interactions types they are interested in.

4 Discussion

The BioPAX format development allowed to standardize and share pathway data with a uniform representation. Although BioPAX is a very rich format allowing to finely describe biological processes, it remains difficult to un-

SIF abstraction rule	Number of nodes	Number of edges	Proportion of edges (%)	Executiontime (s)
controls-state-change-of	416	327	0.54	0.64
controls-transport-of	59	42	0.07	0.13
controls-phosphorylation-of	157	126	0.21	0.15
controls-expression-of	235	247	0.41	0.13
catalysis-precedes	745	16887	27.78	82.37
in-complex-with	5165	19318	31.78	1.17
interacts-with	0	0	0	0.01
neighbor-of	2646	8822	14.51	0.56
consumption-controlled-by	1476	1819	3	0.22
controls-production	1523	1921	3.16	0.26
controls-transport-of-chemical	278	280	0.46	0.09
chemical-affects	1207	1300	2.14	0.38
reacts-with	1081	3956	6.51	0.22
used-to-produce	1546	5746	9.45	70.55
Global SIF abstraction of Reactome	8560	60791		

Table 3: **Number of nodes and edges in the SIF abstractions of Reactome (Homo Sapiens, version 94, September 2025) generated using SPARQL queries**. For each SIF abstraction rule, the number of nodes and edges are presented, as well as the proportion of SIF edge types with regard to the global aggregated SIF network. The total number of nodes and edges for the global SIF abstraction merging the 14 SIF abstractions is also precised.

derstand and to use for non specialist users. Tools and software like Paxtools and ChiBE have been developed to overcome BioPAX' intrinsic complexity and facilitate its use. Among other functions allowing to reduce BioPAX complexity, these tools propose an encapsulated abstraction algorithm for abstracting BioPAX into simpler binary interactions in the SIF format. This abstraction is implemented in Paxtools with encapsulated Java graph patterns that recapitulate SIF abstraction rules. Using these patterns, users can generate a SIF abstraction of a BioPAX model using either the command line tool of Paxtools, its GUI interface named BioPAX-pattern, or the ChiBE visualization software proposing a graphical visualization of the model and the abstraction using the SBGN graphical notation. However, the abstraction rules used that formally describe the BioPAX to SIF abstraction are ambiguously documented at several meanings, making their understanding complicated for the users and potentially leading to misunderstandings of the SIF interactions obtained.

In this article, we investigated the gap between what is presented to the user for describing BioPAX to SIF abstraction rules and what is generated by the SIF Paxtools patterns. We identified three meanings of description of SIF rules (SIF diagram, Paxtools patterns function description and Paxtools patterns code), each one specifying different constraints on the abstraction process (see Figure 2). Beyond the fact that these different meanings of description of SIF rules are ambiguous for users, they reveal a fundamental problem related to the description of the abstraction process on BioPAX models and its reproducibility. Highlighting the heterogeneity of SIF abstraction rules description allows us to demonstrate the importance of the abstraction process' transparency: knowing exactly which entities and relationships are retained or lost during the abstraction process is essential to understand its meaning and evaluate the results. In an effort to highlight the quantitative impact of these different SIF rule descriptions, we compared the results obtained by abstracting a small pathway example from PathBank using three SPARQL queries that recapitulate the constraints of the three meanings of description previously identified (see Figure 3). The number of interactions we obtained was different with each SPARQL query, showing that the different meanings of descriptions of SIF rules can lead to different SIF abstractions and thus a misunderstanding of the abstraction process. Also, we noticed that even state of the art tools that are based on Paxtools patterns for BioPAX to SIF abstraction (Paxtools command line SIF converter, Paxtools GUI interface BioPAX-pattern and ChiBE) do not yield the same results. We obtained zero results with the Paxtools command line SIF converter, which is surprising. Our hypothesis is that it comes from a problem with the mapping of entities to gene identifiers in the Paxtools command line SIF converter. Surprisingly, we found interactions containing ubiquitous molecules (such as water molecules) in the SIF interactions obtained with tools using Paxtools patterns. For example, we retrieved SIF interactions where Aquaporin proteins (Aquaporin-2 and Aquaporin-3) control the transport of water molecules. In the BioPAX model, Aquaporins and water molecules are associated to distinct catalysis and transport events, and no other molecules are involved nor transported. If water is considered as a blacklisted molecule, these interactions should not have matched the abstraction conditions. This raises questions about how Paxtools uses its blacklist of ubiquitous small molecules. Clarification of this point in the Paxtools documentation would be welcome. Our hypothesis is that the updates of Paxtools patterns, including the blacklist updates, could have not been propagated to tools using these patterns.

After identifying these ambiguities in SIF rules descriptions, we developed SPARQL queries for SIF abstraction of BioPAX based on existing SIF abstraction rules. Our goal is to provide transparent, well-defined and modular tools for abstracting BioPAX to SIF. We chose the SPARQL query language to build our SIF abstractions because it is the preferred language for querying data in Semantic Web formats such as BioPAX. Furthermore, SPARQL is a language that complies with FAIR (Findability, Accessiblity, Interoperability and Reusability) and Semantic Web principles, increasingly favored and important for research's reproducibility. We thus developed 14 SPARQL queries for each one of the 14 SIF abstraction rules. We executed our 14 SPARQL queries on two pathway databases BioPAX exports (PathBank and Reactome), demonstrating their efficiency on large scale BioPAX models (see Tables 2 and 3). Our SPARQL queries are independent, they all generate one type of SIF abstraction containing only one SIF edge type. Independent SIF interactions can then be merged to obtain a global SIF abstraction. SPARQL queries respect the modularity of the SIF format by allowing to select the desired interaction types.

We would like to point out that our goal in presenting these SPARQL queries is not to replace Paxtools and ChiBE, which are excellent tools for working with BioPAX models and, to date, the only ones to offer such a wide range of methods for manipulating, processing, and visualizing BioPAX models. Here, we wanted to emphasize the impact of the granularity of SIF rules descriptions on the interpretation of the obtained SIF interactions. In addition, we provided documented SPARQL queries that recapitulate SIF rules in a consistent manner.

5 Conclusion

The study of SIF abstraction rules descriptions revealed a high heterogeneity of documentation. These descriptions span three meanings (SIF rule description from PathwayCommons, Paxtools patterns function description and Paxtools patterns code), the closer to raw code systematically providing additional constraints that were implicit in the other meanings. Using the state of the art tools Paxtools and ChiBE for abstracting BioPAX to SIF, users thus apply patterns that are much more complex than what is described on PathwayCommons. This is an important lack of transparency that can lead to misunderstanding or confusing results. To help clarify these ambiguities and complement the use of Paxtools and Chibe, we propose to represent each of the 14 SIF rule by a SPARQL query based on the standard data schema. The queries are all based on a common meaning of description, making them consistent with each other. SPARQL queries can be used to abstract any BioPAX model to SIF, and work even on large models like pathway databases. These queries are publicly accessible on GitHub and integrated in a Python code to simplify as much as possible their execution and their use across the BioPAX community.

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Declaration on Generative AI

The author(s) have not employed any Generative AI tools.

References

(1) Demir, E. et al. *Nature Biotechnology* **2010**, *28*, Number: 9 Publisher: Nature Publishing Group, 935–942.

(2) Beust, C.; Becker, E.; Théret, N.; Dameron, O. *Computational and Structural Biotechnology Journal* **2024**, *23*, 3999–4010.

(3) Demir, E.; Babur, O.; Rodchenkov, I.; Aksoy, B. A.; Fukuda, K. I.; Gross, B.; Sümer, O. S.; Bader, G. D.; Sander, C. *PLoS Computational Biology* **2013**, *9*, ed. by Prlic, A., e1003194.

(4) Shannon, P.; Markiel, A.; Ozier, O.; Baliga, N. S.; Wang, J. T.; Ramage, D.; Amin, N.; Schwikowski, B.; Ideker, T. *Genome Research* **2003**, *13*, 2498–2504.

(5) Saito, R.; Smoot, M. E.; Ono, K.; Ruscheinski, J.; Wang, P.-L.; Lotia, S.; Pico, A. R.; Bader, G. D.; Ideker, T. *Nature Methods* **2012**, *9*, Publisher: Nature Publishing Group, 1069–1076.

(6) Pundir, S.; Martin, M. J.; O'Donovan, C. *Methods in Molecular Biology (Clifton, N.J.)* **2017**, *1558*, 41–55.

(7) Degtyarenko, K.; de Matos, P.; Ennis, M.; Hastings, J.; Zbinden, M.; McNaught, A.; Alcántara, R.; Darsow, M.; Guedj, M.; Ashburner, M. *Nucleic Acids Research* **2008**, *36*, D344–D350.

(8) Cerami, E. G.; Gross, B. E.; Demir, E.; Rodchenkov, I.; Babur, O.; Anwar, N.; Schultz, N.; Bader, G. D.; Sander, C. *Nucleic Acids Research* **2011**, *39*, D685–690.

(9) Babur, O.; Aksoy, B. A.; Rodchenkov, I.; Sümer, S. O.; Sander, C.; Demir, E. *Bioinformatics (Oxford, England)* **2014**, *30*, 139–140.

(10) Babur, O.; Luna, A.; Korkut, A.; Durupinar, F.; Siper, M. C.; Dogrusoz, U.; Jacome, A. S. V.; Peckner, R.; Christianson, K. E.; Jaffe, J. D.; Spellman, P. T.; Aslan, J. E.; Sander, C.; Demir, E. *Patterns* **2021**, *2*, Publisher: Elsevier, DOI: `10.1016/j.patter.2021.100257`.

(11) Babur, O.; Dogrusoz, U.; Demir, E.; Sander, C. *Bioinformatics (Oxford, England)* **2010**, *26*, 429–431.

(12) Le Novère, N. et al. *Nature Biotechnology* **2009**, *27*, 735–741.

(13) Chang, X.; Yan, S.; Zhang, Y.; Zhang, Y.; Li, L.; Gao, Z.; Lin, X.; Chi, X. *NPJ systems biology and applications* **2024**, *10*, 4.

(14) Kramer, F.; Bayerlová, M.; Klemm, F.; Bleckmann, A.; Beißbarth, T. *Bioinformatics* **2013**, *29*, 520–522.

(15) Wishart, D. S. et al. *Nucleic Acids Research* **2020**, *48*, D470–D478.

(16) Jassal, B. et al. *Nucleic Acids Research* **2020**, *48*, D498–D503.